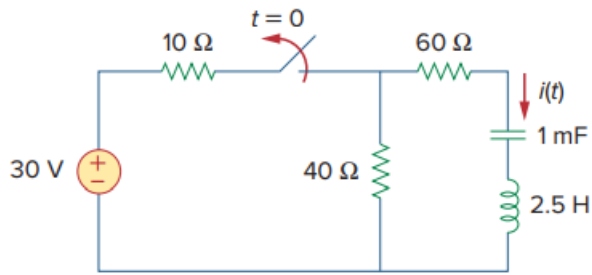


Problem

Find $i(t)$ for $t > 0$ in the circuit of Fig. 8.70.

Figure 8.70:



Step-by-step solution

Step 1 of 9

Consider the following circuit diagram:

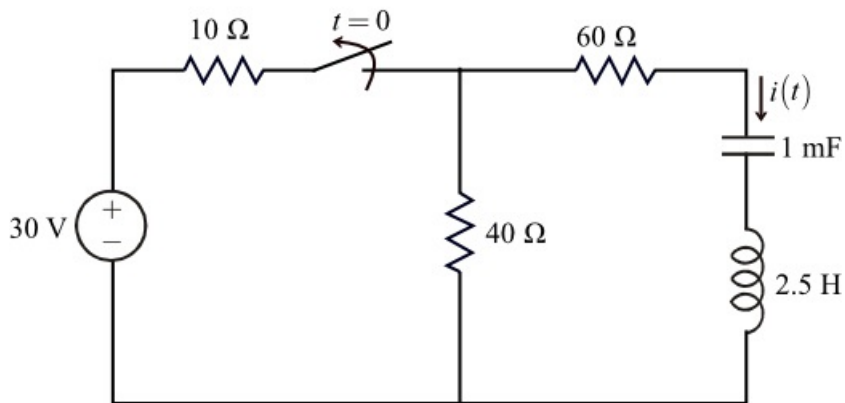


Figure 1

Step 2 of 9

For $t < 0$ the switch is in closed position and circuit acts under steady state condition. At $t = 0^-$ inductor acts like a short circuit and capacitor acts like an open circuit. Hence, modified circuit is shown in Figure 2.

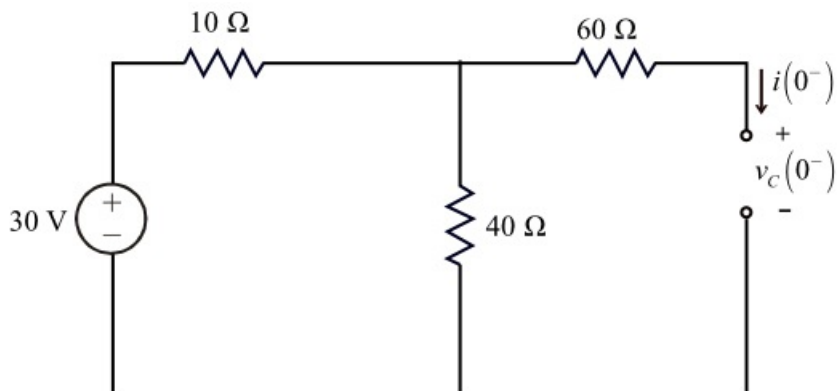


Figure 2

From the Figure 2, the inductor current is,

$$i(0^-) = 0 \text{ A}$$

Calculate the voltage across the open circuit terminals.

$$\begin{aligned} v_c(0^-) &= (30) \left(\frac{40}{40+10} \right) \\ &= (30) \left(\frac{40}{50} \right) \\ &= 24 \text{ V} \end{aligned}$$

Since the inductor current and capacitor voltage cannot change instantaneously.

$$\begin{aligned} v(0) &= v(0^-) \\ &= 24 \text{ V} \end{aligned}$$

And

$$\begin{aligned} i(0) &= i(0^-) \\ &= 0 \text{ A} \end{aligned}$$

Step 4 of 9

For $t > 0$, the switch is opened, therefore the equivalent circuit is shown in Figure 3.

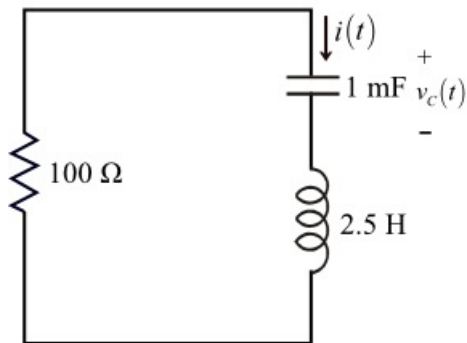


Figure 3

Step 5 of 9

Apply Kirchhoff's voltage law to loop in Figure 3.

$$Ri(t) + v_c(t) + L \frac{di(t)}{dt} = 0 \dots\dots (1)$$

From the figure 3, the current through capacitor is,

$$i(t) = C \frac{dv_c(t)}{dt} \dots\dots (2)$$

Substitute equation (2) in equation (1).

$$\begin{aligned} R \left(C \frac{dv_c(t)}{dt} \right) + v_c(t) + L \frac{d \left(C \frac{dv_c(t)}{dt} \right)}{dt} &= 0 \\ RC \frac{dv_c(t)}{dt} + LC \frac{d^2 v_c(t)}{dt^2} + v_c(t) &= 0 \\ \frac{d^2 v_c(t)}{dt^2} + \frac{R}{L} \frac{dv_c(t)}{dt} + \frac{1}{LC} v_c(t) &= 0 \end{aligned}$$

Calculate the neper frequency (α) from the differential equation.

$$\begin{aligned}\alpha &= \frac{R}{2L} \\ &= \frac{100}{(2)(2.5)} \\ &= 20 \text{ rad/sec}\end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(10^{-3} \times 2.5)}} \\ &= 20 \text{ rad/sec}\end{aligned}$$

Since $\alpha = \omega_0$, the circuit is critically damped. Therefore, the response is,

$$v_C(t) = (A_1 + A_2 t) e^{-20t} \dots\dots (3)$$

Step 7 of 9

Substitute $t = 0$ in equation (3).

$$\begin{aligned}v_C(0) &= (A_1 + A_2(0)) e^{-20(0)} \\ 24 &= A_1 \quad (\text{Since, } v_C(0) = 24 \text{ V}) \\ A_1 &= 24\end{aligned}$$

Differentiate the equation (3) with respect to t .

$$\frac{dv_C(t)}{dt} = -20 A_1 e^{-20t} + A_2 (e^{-20t} - 20t e^{-20t}) \dots\dots (4)$$

Substitute $t = 0$ in equation (4).

$$\begin{aligned}\frac{dv_C(0)}{dt} &= -20 A_1 e^{-20(0)} + A_2 (e^{-20(0)} - 20(0) e^{-20(0)}) \\ \frac{dv_C(0)}{dt} &= -20 A_1 + A_2 \dots\dots (5)\end{aligned}$$

Step 8 of 9

Current through the capacitor at $t = 0$ is,

$$\begin{aligned}i(0) &= C \frac{dv_C(0)}{dt} \\ \frac{dv_C(0)}{dt} &= \frac{i(0)}{C} \\ \frac{dv_C(0)}{dt} &= \frac{0}{C} \\ \frac{dv_C(0)}{dt} &= 0\end{aligned}$$

Step 9 of 9

Substitute 0 for $\frac{dv_c(0)}{dt}$ in equation (5), to obtain the value of A_2 .

$$0 = -20(24) + A_2$$

$$A_2 = 20(24)$$

$$A_2 = 480$$

Therefore, the expression for voltage across capacitor for $t > 0$ is,

$$v_c(t) = (24 + 480t)e^{-20t} \text{ V}$$

Calculate the current through the inductor for $t > 0$.

$$\begin{aligned} i(t) &= C \frac{dv_c(t)}{dt} \\ &= (1 \times 10^{-3}) \frac{d((24 + 480t)e^{-20t})}{dt} \\ &= (1 \times 10^{-3}) \{-480e^{-20t} + 480(e^{-20t} - 20te^{-20t})\} \\ &= (1 \times 10^{-3}) [-9600te^{-20t}] \\ &= -9.6te^{-20t} \text{ A} \end{aligned}$$

Therefore, the expression for current, $i(t)$ for $t > 0$ is $\boxed{-9.6te^{-20t} \text{ A}}$.